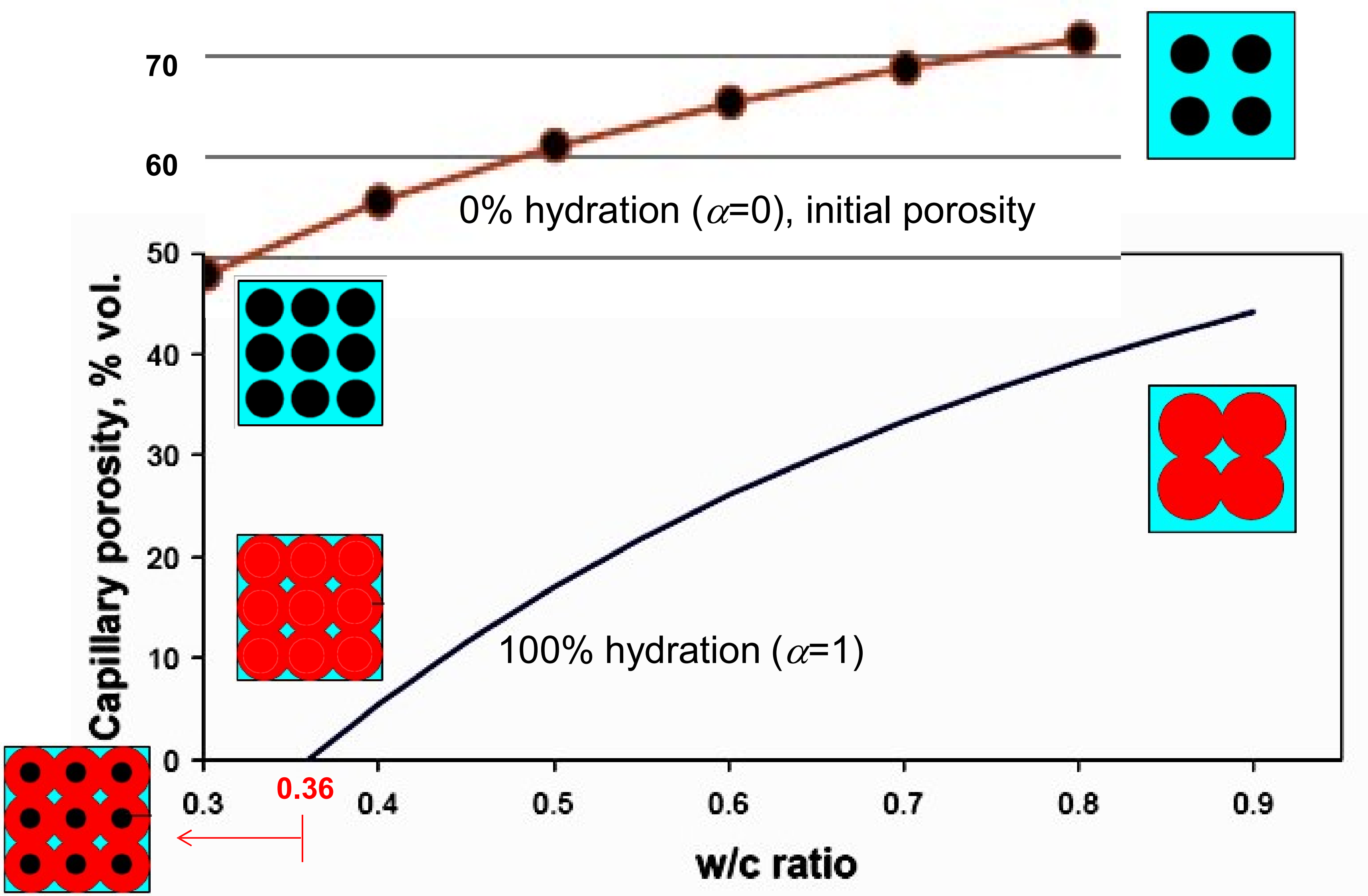
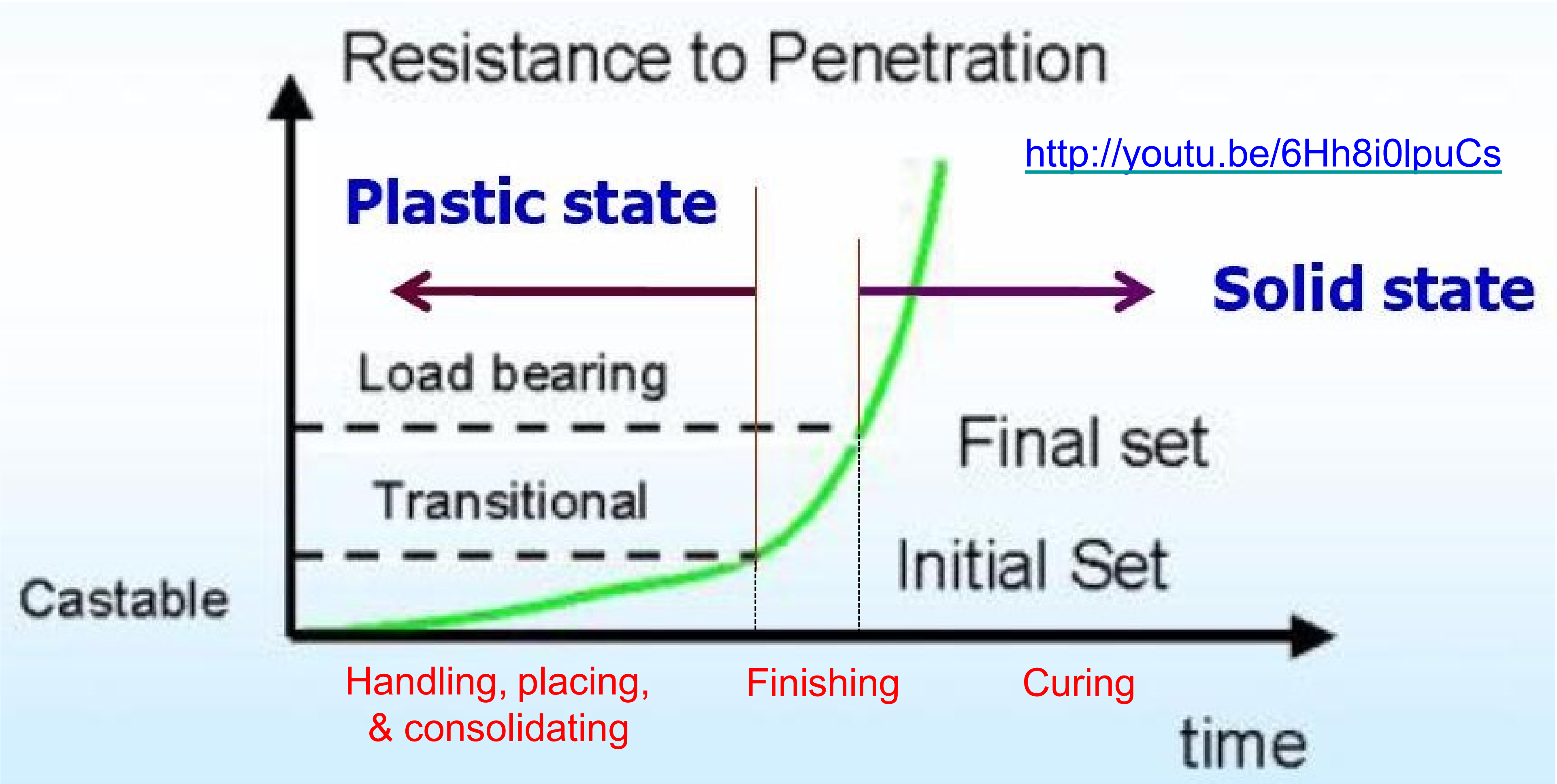
Capillary Porosity as A Function of W/C Ratio and Maturity



Setting of Portland Cement

Setting: stiffening of the cement paste from a fluid to a rigid stage

- Initial set (2 to 4 hrs): paste is beginning to stiffen considerably and can no longer be molded. Handling, placing, & consolidating must be completed before initial set.
- Final set (5 to 8 hrs): paste has hardened to the point at which it can sustain some load. Finishing between initial and final. Curing after final set.



Compressive Strength of Cement

ASTM C109

- Average of three 50 mm **mortar** cubes
- Proportional to compressive strength of cylinders
- Compressive strength of concrete cannot be accurately predicted from cement strength

Mold

Prepare sample

Compression test

Typical failure

http://youtu.be/gX4dD0li38I

Standard Portland Cement Types (ASTM C 150)

- Type I: Normal
- Type II: Moderate Sulfate Resistance/Heat of Hydration
- Type III: High Early Strength
  - Becoming cheaper & more common
  - We can strip forms earlier and speed up production
- Type IV: Low Heat of Hydration
  - For large, massive pours to control heat of hydration
- Type V: High Sulfate Resistance

Typical Chemical Composition and Properties of Portland Cements, ASTM Types I to V

ASTM Type[a]	I	II	III	IV	V
C_3S	55	55	55	**42**	55
C_2S	18	19	17	32	22
C_3A	10	**6**	10	**4**	**4**
C_4AF	8	11	8	15	12
$C\underline{S}H_2$	6	5	6	4	4
Fineness (Blaine, m^2/kg)	365	375	**550**	340	380
Compressive strength[b] (1 day, MPa)	15	14	**24**	**4**	12
Heat of Hydration (7 days, J/g)	350	265	370	**235**	310

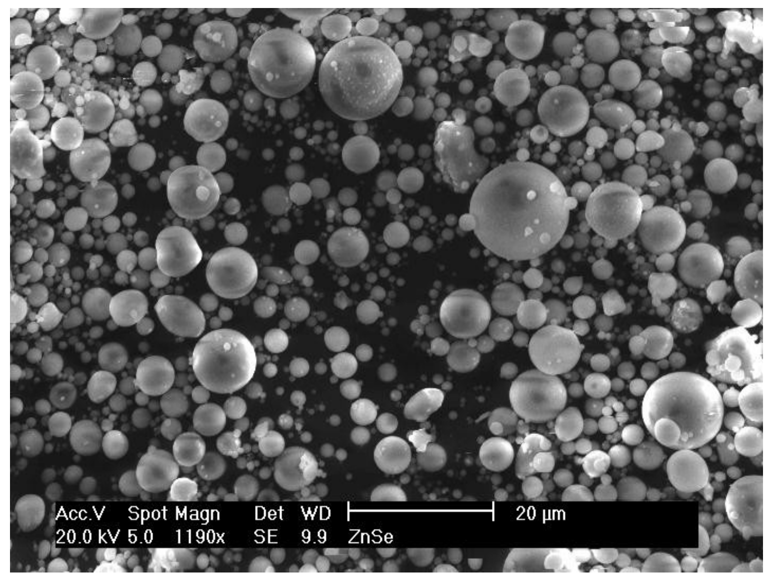
[a] Canadian Standards Association designations are 10, 20, 30, 40, and 50, respectively
[b] Test on 50 mm mortar cubes (ASTM C 109)

Supplementary Cementitious Materials (SCM)

- Inorganic material
- Alternative names: *Cement replacement materials, cement substitutes, mineral admixtures*
- Particle size similar or smaller than that of cement
- Used as either partial cement replacement (or addition)
- Added separately as concrete ingredient
- Incorporated in cement (Blended cement)
- Common SCM
    - Fly ash
    - Silica fume
    - Ground granulated blast-furnace slag

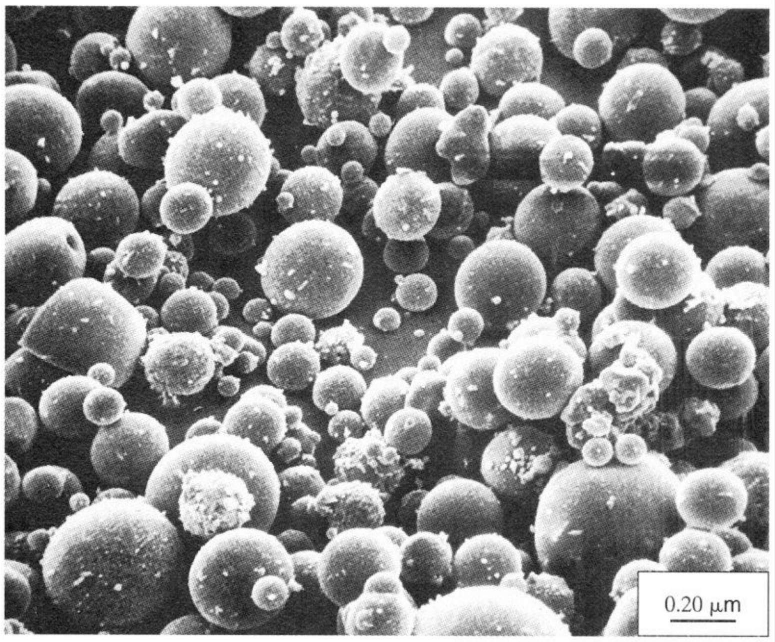
Fly Ash

- By-product of coal combustion in power stations
- Most commonly used SCM in civil engineering structures
- Particle size is similar to cement
- Two types of fly ash (ASTM C 618)
  - Class C fly ash: pozzolanic and cementitous
  - Class F fly ash: pozzolanic
- http://youtu.be/wwM880Savhg



Silica Fume

- By-product from in the manufacture of silicon metal and alloys
- Super Pozzolan
  - High fineness (particle size is 1/100 of cement)
  - High silica content (>95%)
- Challenge on handling
  - Wet: slurry in water
  - Dry: densified
    & undensified form
    - Challenge on dispersion

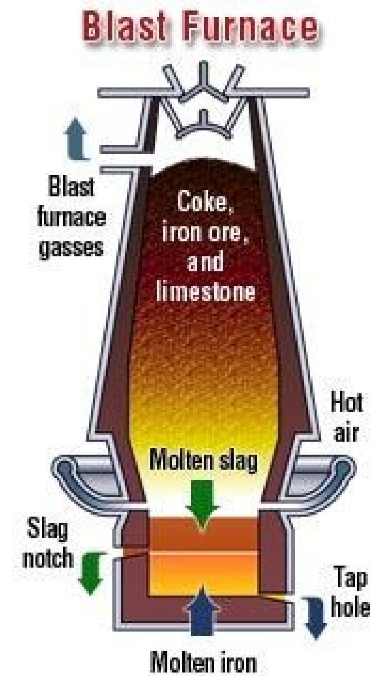


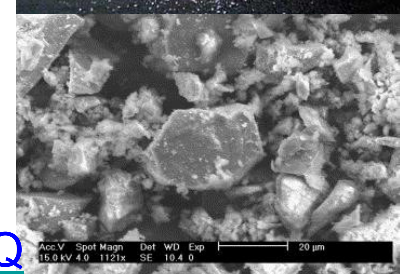
Pozzolanic Reaction

- Portland cement
  - $C_3S + H \xrightarrow[\text{fast}]{} C\text{-}S\text{-}H + CH$
- Pozzolanic reaction
  - $CH + \text{pozzolan (S)} + H \xrightarrow[\text{slow}]{} C\text{-}S\text{-}H$
- Addition of pozzolan has similar effect to raising C_2S content of cement (i.e. lower heat evolution and lower early strength)
- Slow reaction requires prolonged moist curing; otherwise pozzolan act mainly as a filler

Blast-furnace Slag

- Residues from metallurgical processes, i.e. the blast furnace production of ***iron*** from ore (rich in lime, silica, and alumina)
- Slag must be rapidly cooled (quenched) to form a hydraulically active calcium aluminosilicate glass
- The smaller granules (< 4 mm) are ground for use as a mineral admixture, i.e. ground granulated blast-furnace slag (GGBS)
- The larger pellets, which are porous and partially crystalline, can be used as a lightweight aggregate





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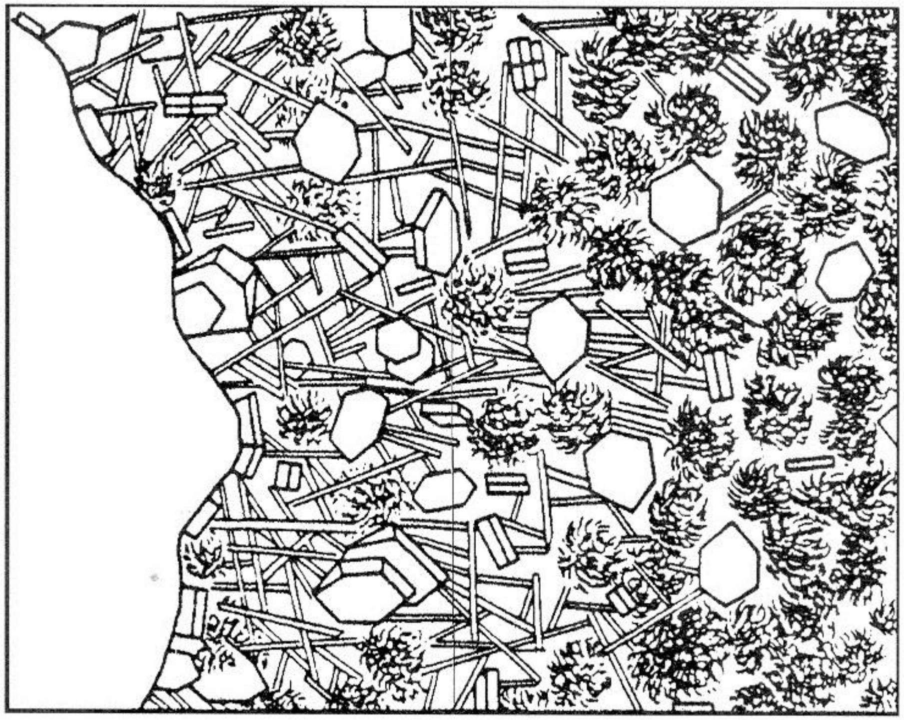
Slag Hydration

- Slag reacts slowly with water due to the presence of impervious coatings of amorphous silica and alumina that form around slag particles early in the hydration process
- Slag needs to be activated by alkaline compounds
- Slags are most commonly activated by Portland cement, where $Ca(OH)_2$ formed during hydration is the principal activator. Only 10-20% of cement is needed for activation
- The rate of hydration of activated slag is slow, similar to that of C_2S, as is the heat of hydration

SCM …

- Reduce overall heat of hydration & thermal cracking
- Reduce porosity & permeability of concrete
- Increase ultimate strength of concrete
- Improve durability of concrete
- Improve workability (consistency and cohesiveness)
- Is more economic and sustainable

Interfacial Transition Zone



- A thin zone surrounding aggregate particles in which structure of cement paste is different from that of bulk paste farther away from the physical interface, in terms of density, morphology, and composition.
- Due to inability of cement particles to pack efficiently around aggregates (wall effect). This raises the local w/c ratio, which can be further increased by localized bleeding

Pozzolanic reaction increases:

(1) C-S-H content and binder paste strength

(2) improve aggregate-cement paste bond strength

(3) reduces the porosity and modifies pore-size distribution

SCM and Sustainability

One Tonne of Portland Cement

- *1.7 Tonne of Raw Materials required*
- *4000 to 7500 MJ Energy consumed*
- *1 tonne of Carbon Dioxide liberated*

One Tonne of Supplementary Cementitious Materials (SCM) production requires:

- *700 to 1000 MJ for Slag*
- *150 to 400 MJ for Fly Ash*